

Part 1: Report

Securing a system using a packet filtering firewall.

Firewalls is a security system that acts as a barrier surrounding a machine that can monitor and control network traffic, allowing only trusted or authorised networks from entering or leaving. Two protocols are SSH and HTTP. SSH or Secure Shell is a cryptographic network protocol which is used for secure remote access to systems over an unsecured network. HTTP or Hypertext Transfer Protocol is a protocol used for sending content or information over the internet.

```
(kali@ovmsdevcg01q)-[~]
$ cat /etc/services | grep ssh
ssh                22/tcp            # SSH Remote Login Protocol

(kali@ovmsdevcg01q)-[~]
$ /etc/init.d/ssh start
Starting ssh (via systemctl): ssh.service.
```

Checking that ssh is installed and then starting it

```
(kali@ovmsdevcg01q)-[~]
$ cat /etc/services | grep http
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
http              80/tcp            www               # WorldWideWeb HTTP
https             443/tcp           # https protocol over TLS/SSL
https             443/udp           # HTTP/3
http-alt          8080/tcp          webcache          # WWW caching service

(kali@ovmsdevcg01q)-[~]
$ /etc/init.d/apache2 start
Starting apache2 (via systemctl): apache2.service.
```

Checking that http is installed and then starting it

```
(kali@ovmsdevcg01q)-[~]
$ sudo systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
... skipping ...
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
```

This show that ssh wasn't initially running

```
(kali@ovmsdevcg01q)-[~]  
$ sudo apt-get install iptables  
[sudo] password for kali:  
Reading package lists ... Done  
Building dependency tree ... Done  
Reading state information ... Done  
iptables is already the newest version (1.8.9-2).  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Installing iptables

1. Configure your firewall to do the following:

a. Reject all SSH packets.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables -A INPUT -p tcp --dport 22 -j REJECT
```

- The above rule prevents anyone using ssh or secure shell packets from connecting

b. Allow SSH remote connections.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

- The above rule allows people to connect using ssh

c. Deny ping.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

- The above rule denies ping making the machine harder to reach or connect to

d. Reject all traffic coming to port 80.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables -A INPUT -p tcp --dport 80 -j REJECT
```

- The above rule prevents any incoming traffic from reaching port 80 or http

e. Block incoming traffic connection to the IP address of your virtual machine.

```
(kali@ovmsdevcg01q)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether aa:c2:13:7a:5d:2f brd ff:ff:ff:ff:ff:ff
    inet 192.168.64.3/24 brd 192.168.64.255 scope global dynamic noprefixroute eth0
        valid_lft 85122sec preferred_lft 85122sec
    inet6 fd26:6f1f:53b4:f05e:d3b:abf3:2d1d:90e/64 scope global temporary dynamic
        valid_lft 603525sec preferred_lft 84690sec
    inet6 fd26:6f1f:53b4:f05e:a8c2:13ff:fe7a:5d2f/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591891sec preferred_lft 604691sec
    inet6 fe80::a8c2:13ff:fe7a:5d2f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Here we found the ip address of our virtual machine which i have highlighted

```
(kali@ovmsdevcg01q)-[~]
$ sudo iptables -A INPUT -s 192.168.64.3 -j DROP
```

- The above rule is essentially an extra layer of security which blocks incoming connections from targeting the IP address of our virtual machine. An IP address is a unique identifier given to all devices which computers read, it's how devices communicate with each other.

f. Allow traffic coming to port 80 (inbound) but reject traffic going out (outbound) through port 80.

```
(kali@ovmsdevcg01q)-[~]
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

(kali@ovmsdevcg01q)-[~]
$ sudo iptables -A OUTPUT -p tcp --dport 80 -j REJECT
```

- The above rule allows connections to http, but stops connections from leaving through http.

2.Record all firewall rules and results in the report.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables-save > FireWallRules.text
```

We saved the rules we made in question 1 which are above in a file FireWallRules.text

```
(kali@ovmsdevcg01q)-[~]  
$ cat FireWallRules.text  
# Generated by iptables-save v1.8.9 (nf_tables) on Wed May 31 10:40:20 2023  
*filter  
:INPUT ACCEPT [0:0]  
:FORWARD ACCEPT [0:0]  
:OUTPUT ACCEPT [0:0]  
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT  
-A INPUT -p icmp -m icmp --icmp-type 8 -j DROP  
-A INPUT -p tcp -m tcp --dport 80 -j REJECT --reject-with icmp-port-unreachab  
le  
-A INPUT -s 192.168.64.3/32 -j DROP  
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT  
-A INPUT -p tcp -m tcp --dport 22 -j REJECT --reject-with icmp-port-unreachab  
le  
-A OUTPUT -p tcp -m tcp --dport 80 -j REJECT --reject-with icmp-port-unreacha  
ble  
COMMIT  
# Completed on Wed May 31 10:40:20 2023
```

Using the 'cat' command we can view our rules.

```
(kali@ovmsdevcg01q)-[~]  
$ sudo iptables -L --line-number  
[sudo] password for kali:  
Chain INPUT (policy ACCEPT)  
num target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
num target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
num target      prot opt source                destination
```

We can also see our firewalls through the above command.

Different firewall rules like the ones above have individual purposes which contribute to better control network traffic based on specific criteria or rules. The images show some different rules that filter incoming and outgoing traffic. Some of the keywords that allow this are accept, reject and drop.

Accept is when a firewall rule where if the traffic is matching the rule it is then allowed to pass through the firewall. Reject is when a firewall rule that if the traffic is matching that rule it is then

explicitly denied. Also When traffic is rejected, the firewall sends a rejection message back to the source, indicating that the requested connection is not allowed, helping to inform the source. Drop is when a firewall rules that if the traffic is matching the rule the traffic is then discarded or dropped. This differs from reject where a message is sent back to the source

3. Describe in detail how you tested all configurations with real practical tests and/or with your gathered information in the report.

We are using the MAC terminal to test our firewalls made above

ovmsdevcg01q 192.168.64.255

- This is the hostname and ip address of my virtual machine which is used below to connect my terminal with the virtual machine

Fire wall A

```
[aidenaiden@Aidens-MacBook-Air ~ % nc -zv ovmsdevcg01q 192.168.64.255 22 ]
ssh: connect to host 192.168.64.255 port 22: Operation timed out
```

- As connection timed out it shows that connection through ssh are blocked

Fire wall B

```
[aidenaiden@Aidens-MacBook-Air ~ % nc -zv ovmsdevcg01q 192.168.64.255 22 ]
```

- Refer to firewall A

Fire wall C

```
[aidenaiden@Aidens-MacBook-Air ~ % ping 192.168.64.255
PING 192.168.64.255 (192.168.64.255): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
Request timeout for icmp_seq 4
Request timeout for icmp_seq 5
Request timeout for icmp_seq 6
Request timeout for icmp_seq 7
Request timeout for icmp_seq 8
Request timeout for icmp_seq 9
Request timeout for icmp_seq 10
Request timeout for icmp_seq 11
Request timeout for icmp_seq 12
Request timeout for icmp_seq 13
Request timeout for icmp_seq 14
Request timeout for icmp_seq 15
```

Fire wall D

```
[aidenaiden@Aidens-MacBook-Air ~ % nc -zv ovmsdevcg01q 192.168.64.255 80
```

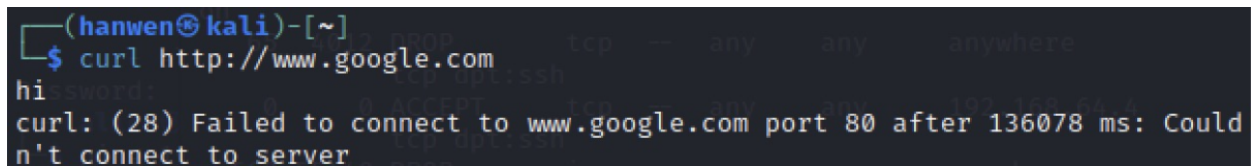
http: connect to host 192.168.64.255 port 80: Operation timed out

- The connection through the my mac and the virtual machine using http or port 80 failed showing that the firewall is working

Fire wall E

- With prior rules already preventing connections to the virtual machine they will already not be able to access the ip address

Fire wall F



```
(hanwen@kali)-[~]  
$ curl http://www.google.com  
hi  
curl: (28) Failed to connect to www.google.com port 80 after 136078 ms: Could  
n't connect to server
```

- curl is a function on kali linux that is used for making HTTP requests to servers, in this case Google, in which it failed to connect, showing that the firewall rule that rejects http connection is working.

4. Discuss the advantages and disadvantages of firewalls with iptables and make suggestions to overcome the disadvantages in your report.

Advantages of Firewalls with iptables:

One advantage of fire walls is with the enhanced Security, firewalls with iptables are structured in a way that they act as a single choke point, keeping unauthorised individuals out of a network by filtering incoming and outgoing traffic based on predefined rules. This significantly reduces the attack surface and helps protect sensitive data and resources. A second advantage is with the centralised security monitoring, where by implementing a firewall, organisations can establish a centralised location for monitoring security-related threats. This allows for better identification of potential threats, and a faster response to security attacks. Another advantage is with its multipurpose functionality, this is where Firewalls with iptables offer a convenient platform for various internet functions beyond security. They can be configured to perform tasks such as network address translation (NAT), load balancing, and traffic shaping, providing additional value and flexibility to the network infrastructure.

Disadvantages of Firewalls with iptables:

One disadvantage is with the limitations in attack protection, while firewalls with iptables are relatively effective at blocking and filtering network traffic, they cannot protect against all types of

attacks. Sophisticated attacks such as those that bypass firewall rules or use encrypted channels can still pose a threat to the network. A second advantage is incomplete Internal Threat Protection, where firewalls primarily focus on securing external connections, but they may not provide full protection against internal threats. Insider attacks or compromised internal systems can bypass firewall defences, emphasising the need for additional security measures such as access controls and intrusion detection systems within the network. Another limitation is with the potential vulnerabilities with Wireless LAN, Improperly secured wireless local area networks (LANs) can be accessed from outside the network perimeter, bypassing the firewall's protection. Attackers within range of the wireless network may exploit weaknesses in the wireless security protocols or gain unauthorised access to sensitive data. Another disadvantage could be with the external threats. firewalls may not directly protect against threats introduced through external devices such as USB devices. If an infected device is connected to a network, it can potentially introduce malware or compromise network security, requiring additional measures such as endpoint protection and device management.

Suggestions to overcome disadvantages

A way to overcome the limitations in attack protection provided by firewalls, could be implementing multiple layers of security. This means combining the use of firewalls with other security measures. Some security measures could be something like regular security audits. In having multiple layers of defence, we can better protect against a wide range of attack techniques and minimise the chances of successful breaches or damage to a system. A way to address the issue of incomplete internal threat protection, could be by implementing user authentication mechanisms. This means ensuring that only authorised individuals have access to interact or see sensitive resources and systems related to the firewalls. Additionally, these mechanisms can help detect and prevent insider threats by monitoring and analysing user behaviour for any malicious activities. A way to minimise the vulnerabilities in wireless LAN, is to regularly update the system. Updates usually include security patches that fix known vulnerabilities. By keeping the wireless access points up to date, we can reduce the risk of unauthorised users having access to the network through wireless connections. A way to protect against external threats from devices such as USB drives we can teach employees about the risks and promote safe practices, whilst ensuring and enforcing rules. This includes regularly training employees who interact with firewalls or iptables on the importance of not connecting untrusted devices to the network. By training or upholding procedures promoting responsible device usage, we can minimise the potential threats of infiltration.

5.Discuss the roles and significance of packet filtering, circuit relay and application layer firewalls in building multilayer firewalls. Comment on how the use of multilayer firewalls impacts on the network traffic delivery time.

Packet Filtering is the most basic and operates at the network layer (Layer 3) of the OSI model in which it filters incoming and outgoing network packets based on predetermined rules that are

set, this can be shown with the rules made earlier in the report (Stallings W 2017, Network security essentials: applications and standards, 6th edn, Pearson, Essex). These rules typically consider information such as source and destination IP addresses, port numbers, and protocol types. Packet filtering firewalls allow or block packets based on these rules, creating a barrier between trusted and untrusted networks. However packet filtering firewalls lack the ability to inspect the contents of the packets beyond basic header information.

Circuit relays operate at the transport layer (Layer 4) of the OSI model. In which they examine header information similar to packet filtering firewalls, they maintain a state table that tracks the state of network connections. This includes information about established connections, such as TCP sessions. By keeping track of the connection state, circuit relays help make better decisions in regards to allow or block traffic. Circuit relay firewalls can dynamically adjust their filtering rules based on the specific state of connections, which upholds a higher level of control over network traffic.

Application Layer firewalls, which is also known as a proxy firewall, operates at the application layer (Layer 7) of the OSI model. It provides advanced inspection and control where it acts as a connection between client devices and remote servers. When a client requests access to a remote application or service, the application level gateway establishes a separate connection with the server on behalf of the client. It inspects and filters application-layer traffic, allowing a level of control over the types of applications and services that could be accessed. The gateway also provides additional security features such as authentication and content filtering.

Multilayer firewalls are the combination of different types of firewalls such as the ones mentioned above in which they help to provide a strong but layered approach to network security. By deploying multiple layers of firewalls, it can contribute to organisations as with it it comes with the advantages of each of the layers, strengthening the overall security posture. The use of multi layer firewalls however has its limitations, this is where it can impact network traffic delivery time, making connection slower. Each layer of the firewall introduces additional processing and inspection of packets, which can increase latency and delays in sending network traffic. The impact on network traffic delivery time further depends on the complexity of firewall rules where more complex will take longer despite being stronger.

overall multilayer firewalls provide more secure security by leveraging the strengths of packet filtering, circuit relay, and application layer firewalls as well as others. While they enhance network security, they take longer where individuals should consider the balance in security requirements and network performance to ensure we maintain the fast and efficient traffic delivery.

In conclusion, securing a system using a packet filtering firewall is necessary in order to protect network resources and data, whether they be information from our local computers or large organisations. Above we discussed the installation and configuration of SSH and HTTP protocols, the implementation of iptables for firewall management, as well as the testing of firewall rules through practical tests to ensure our codes work. With firewalls we unpacked the advantages and disadvantages of firewalls with iptables and suggested measures to overcome

limitations. The roles and importance of packet filtering, circuit relay as well as application layer firewalls in building multilayer firewalls were also explained. By considering the recommendations and maintaining a balance between security and network performance, organisations can establish secure and safe firewall systems to safeguard against potential threats.

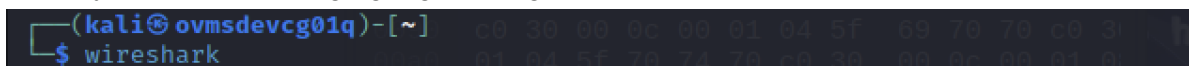
Part 2 below

Part 2: Network traffic analysis

- Packet analysis using Wireshark

1. Analysis of the web page google. using wireshark packet capture (HTTP)

```
(kali@ovmsdevcg01q)-[~]  
$ wireshark
```



Starting up wireshark (RMIT/ Lab Manual Intro to Cybersecurity ver 1.1.pdf)

```
(kali@ovmsdevcg01q)-[~]
$ wget http://www.google.com
--2023-06-09 09:06:25-- http://www.google.com/
Resolving www.google.com (www.google.com)... 142.250.67.4, 2404:6800:4006:811
::2004
Connecting to www.google.com (www.google.com)|142.250.67.4|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html [ => 142.250.67.4 ] 17.92K --.-KB/s in 0.02s

2023-06-09 09:06:25 (788 KB/s) - 'index.html' saved [18351]
```

Use of 'wget' to connect to google as a http so that we can later filter for the http(Lab Manual Intro to Cybersecurity1.1.pdf)

a.3 different protocols that appear in the protocol column in the unfiltered packet-listing window.

3	0.026784213	192.168.64.1	192.168.64.3	DNS	90	Standard query response
4	0.028712257	192.168.64.1	192.168.64.3	DNS	102	Standard query response
5	0.029491009	192.168.64.3	142.250.67.4	TCP	74	48654 → 80 [SYN] Seq=0
6	0.057755675	142.250.67.4	192.168.64.3	TCP	74	80 → 48654 [SYN, ACK] s
7	0.057846518	192.168.64.3	142.250.67.4	TCP	66	48654 → 80 [ACK] Seq=1
8	0.057991784	192.168.64.3	142.250.67.4	HTTP	195	GET / HTTP/1.1

- TCP or Transmission Control Protocol is a method used for sending data securely over networks. It ensures that data is transmitted correctly and in the right order.
- DNS or Domain Name System is a protocol that converts domain names into IP addresses. This helps us access websites where it converts human-readable names into computer-readable addresses.
- HTTP or Hypertext Transfer Protocol is a protocol that operates to allow web browsers to communicate with web servers. It is the basis of the internet where it allows the retrieval of web content, such as web pages, images, and videos online.

b.The time it took from when the HTTP GET message was sent until the HTTP OK reply was received

No.	Time	Source	Destination	Protocol	Length	Info
8	0.057991784	192.168.64.3	142.250.67.4	HTTP	195	GET / HTTP/1.1
22	0.278573133	142.250.67.4	192.168.64.3	HTTP	2796	HTTP/1.1 200 OK (text/html)

To analyse the HTTP get and HTTP ok we filtered through the packets using http as shown above. It took 0.22058135 milliseconds, this is calculated through finding the difference between the HTTP get and the HTTP ok. The time is written under the time column and we subtracted 0.278573133 with 0.057991784.

c.The IP address of the website and the IP address of your Kali Linux VM

Source	Destination
192.168.64.3	142.250.67.4
142.250.67.4	192.168.64.3

Looking specifically at the destination and source columns we see the ip addresses, the destination for the HTTP get is google's ip address, thus it is 142.250.67.4 leaving our own virtual machines ip address of 192.168.64.3.

2. Analysis of the web page google. using wireshark packet capture looking at ping (ICMP)

```
(kali@ovmsdevcg01q)-[~]
$ ping -c 5 www.google.com
PING www.google.com (142.250.67.4) 56(84) bytes of data.
64 bytes from syd15s16-in-f4.1e100.net (142.250.67.4): icmp_seq=1 ttl=114 time=36.9 ms
64 bytes from syd15s16-in-f4.1e100.net (142.250.67.4): icmp_seq=2 ttl=114 time=28.6 ms
64 bytes from syd15s16-in-f4.1e100.net (142.250.67.4): icmp_seq=3 ttl=114 time=37.1 ms
64 bytes from syd15s16-in-f4.1e100.net (142.250.67.4): icmp_seq=4 ttl=114 time=36.8 ms
64 bytes from syd15s16-in-f4.1e100.net (142.250.67.4): icmp_seq=5 ttl=114 time=31.0 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4015ms
rtt min/avg/max/mdev = 28.646/34.099/37.122/3.556 ms
```

a. Analysis of The ping request packets sent by VM

No.	Time	Source	Destination	Protocol	Length	Info
5	0.002045396	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id=0xf8
6	0.038876899	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id=0xf8
9	1.006220591	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id=0xf8
10	1.034791096	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id=0xf8
13	2.009011030	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id=0xf8
14	2.046036245	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id=0xf8
17	3.012978568	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id=0xf8
18	3.049723371	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id=0xf8
21	4.017145045	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id=0xf8
22	4.048138574	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id=0xf8

Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x57ba [correct]
[Checksum Status: Good]
Identifier (BE): 63517 (0xf81d)
Identifier (LE): 7672 (0x1df8)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[Response frame: 6]
Timestamp from icmp data: Jun 9, 2023 09:29:28.
[Timestamp from icmp data (relative): 0.42884064]
Data (48 bytes)

The ICMP type is 8 and the ICMP code number is 0. ICMP type 8 represents an Echo Request, which is used to initiate a request for a network device (typically a computer or network host) to send an Echo Reply back. The ICMP code 0, in this case, indicates that it is a general Echo Request message without any specific code associated with it.

We can check for other fields by seeing which ones have a byte, all but checksum status are fields. The number of bytes for the checksum, sequence number and identifier fields are all two bytes, this can be seen in the image above on the bottom.

b. Analysis of The ping reply packets sent by VM

No.	Time	Source	Destination	Protocol	Length	Info
5	23:29:28.428840648	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id
6	23:29:28.465672151	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id
9	23:29:29.433015843	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id
10	23:29:29.461586348	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id
13	23:29:30.435806282	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id
14	23:29:30.472831497	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id
17	23:29:31.439773820	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id
18	23:29:31.476518623	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id
21	23:29:32.443940297	192.168.64.3	142.250.67.4	ICMP	98	Echo (ping) request id
22	23:29:32.474933826	142.250.67.4	192.168.64.3	ICMP	98	Echo (ping) reply id

Internet Protocol Version 4, Src: 142.250.67.4, Ds	0000	aa	c2	13	7a	5d	2f	1a	3e	ef	1e	8e	64	08	00	45
Internet Control Message Protocol	0010	00	54	00	00	00	00	72	01	75	ff	8e	fa	43	04	c0
Type: 0 (Echo (ping) reply)	0020	40	03	00	00	5f	ba	f8	1d	00	01	58	64	82	64	00
Code: 0	0030	00	00	08	8b	06	00	00	00	00	00	10	11	12	13	14
Checksum: 0x5fba [correct]	0040	16	17	18	19	1a	1b	1c	1d	1e	1f	20	21	22	23	24
[Checksum Status: Good]	0050	26	27	28	29	2a	2b	2c	2d	2e	2f	30	31	32	33	34
Identifier (BE): 63517 (0xf81d)	0060	36	37													
Identifier (LE): 7672 (0x1df8)																
Sequence Number (BE): 1 (0x0001)																
Sequence Number (LE): 256 (0x0100)																
[Request frame: 5]																
[Response time: 36.832 ms]																
Timestamp from icmp data: Jun 9 2023 09:29:28																

Identifier (little endian repr...tion) (icmp.ident_le), 2 byte

ICMP is a network protocol that helps devices communicate and diagnose network issues, while ping is a simple tool that uses ICMP to check if a device is reachable and measure the response time. So if ping is slow it correlates to network issues. The ICMP type is 0 and the ICMP code number is also 0 for the ping reply. This means the ICMP type 0 is a response sent by a device to acknowledge that it has received an ICMP Echo Request. The code 0 indicates that it is a general Echo Reply message without any specific additional code.

The following are fields where they contain at least one byte, type:0, code: 0, checksum: 0x5fba, identifier: (BE), identifier (LE), sequence Number (BE), sequence Number(LE), timestamps and data. And the number of bytes for the checksum, sequence number and identifier fields are all two bytes.

c. Comparing the RTT time reported in the command window with the timestamps on Wireshark

```

— www.google.com ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4015ms
rtt min/avg/max/mdev = 28.646/34.099/37.122/3.556 ms

```

We can look at the RTT time by looking at the minimum which is 28.646 and the max 37.122.

Then we can check the time stamps on wireshark to check whether they are in between the min and max values. (Ikram UI Haq/ class workshop/ 8th june/assignment help)

[Response time: 36.832 ms]

[Response time: 28.571 ms]

This is the response time on wireshark and as we can see it is in between the min and max values

3. Analysis of the web page google. using wireshark packet capture looking at domain name (DNS)

DNS or domain name system translates domain names such as google.com and converts it into IP addresses so that computers can communicate with each other over the internet. (Ikram UI Haq/ class workshop/ 8th june/assignment help)

```
(kali@ovmsdevcg01q)-[~]
$ nslookup www.google.com
Server:      192.168.64.1
Address: 192.168.64.1#53
Non-authoritative answer: Dst Port:
Name:   www.google.com
Address: 142.250.67.4
Name:   www.google.com
Address: 2404:6800:4006:811::2004
```

No.	Time	Source	Destination	Protocol	Length	Info
1	00:07:52.377942767	192.168.64.3	192.168.64.1	DNS	74	Standard query 0x8c85 A www.google.com
4	00:07:52.398511845	192.168.64.1	192.168.64.3	DNS	90	Standard query response 0x8c85 A www.google.com
5	00:07:52.398838719	192.168.64.3	192.168.64.1	DNS	74	Standard query 0x353a AAAA www.google.com
6	00:07:52.442140115	192.168.64.1	192.168.64.3	DNS	102	Standard query response 0x353a AAAA www.google.com

▶ Ethernet II, Src: aa:c2:13:7a:5d:2f (aa:c2:13:7a:5d:2f), Dst: 08:00:00:00:00:00	0000	1a 3e ef 1e 8e 64 aa c2 13 7a 5d 2f 08 00 45 00
▶ Internet Protocol Version 4, Src: 192.168.64.3, Dst: 192.168.64.1	0010	00 3c e9 95 00 00 40 11 8f c6 c0 a8 40 03 c0 00
▶ User Datagram Protocol, Src Port: 38538, Dst Port: 53	0020	40 01 96 8a 00 35 00 28 4c 5c 8c 85 01 00 00 00
▶ Domain Name System (query)	0030	00 00 00 00 00 00 03 77 77 77 06 67 6f 6f 67 6f
	0040	65 03 63 6f 6d 00 00 01 00 01

Domain Name System: Protocol Packets: 6 · Displayed: 4 (66.7%) · Dropped: 0 (0.0%) Profile: Default

a. Analysis of UDP in wireshark

UDP is a communication protocol that allows data to be sent between devices over a network without establishing a direct connection. For the analysis of google through filtering using the UDP and analyzing one there are 5 fields this is where they each contain at least one byte:

- Source port: 38538
- Destination port: 53
- Length: 40
- Checksum: 0x4c5c
- UDP payload

The RFC for the UDP is none of the header fields

b. The length (in bytes) of each of the UDP header fields.

- Source port: 38538 has 2 bytes
- Destination port: 53 has 2 bytes

- Length: 40 has 2 bytes
- Checksum: 0x4c5c has 2 bytes
- UDP payload has 32 bytes

References

Ikram UI Haq/ class workshop/ 8th june/assignment help

Stallings W 2017, Network security essentials: applications and standards, 6th edn, Pearson, Essex

RMIT/ Lab Manual Intro to Cybersecurity ver 1.1.pdf